

# Inventory ABC Classification, EOQ & Reorder Point

Findings Report — Online Retail Dataset (3,801 SKUs)

Daniel Ricardo Forero Guerrero · Industrial Engineering, Universidad El Bosque · September 2026

---

## The problem

A retailer with thousands of SKUs cannot manage every product with the same level of attention. Spending equal effort monitoring a low-volume item and a top seller wastes time and ties up capital in the wrong places. This project classifies 3,801 products by sales value, calculates an order quantity and reorder point for each, and flags the products that most need close monitoring.

## Methodology

|                                        |                                                                                                                                                                                                                                      |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ABC Classification</b>              | Products ranked by annual sales value; cumulative share of value determines category (A: up to 80%, B: 80–95%, C: 95–100%). The 80% threshold is an industry convention, not a value derived from the data.                          |
| <b>Economic Order Quantity (EOQ)</b>   | $EOQ = \sqrt{2 \times \text{Annual Demand} \times \text{Order Cost} / \text{Holding Cost}}$ . Order cost (£15/order) and holding cost (20% of unit price) are documented assumptions, not present in the raw data.                   |
| <b>Reorder Point (ROP)</b>             | $ROP = (\text{Daily Demand} \times \text{Lead Time}) + \text{Safety Stock}$ , with Safety Stock at a 95% service level ( $1.65 \times \text{Std. Dev.} \times \sqrt{\text{Lead Time}}$ ). Lead time: 7 days (documented assumption). |
| <b>Demand Variability (CV) Ranking</b> | Coefficient of Variation ( $\text{Std. Dev.} / \text{Daily Demand}$ ) ranks Category A products by how unpredictable their demand is, independent of sales volume — used to build the Top 20 risk watch-list.                        |

## Key findings

- **21.44% of products (Category A, 815 SKUs) generate 79.99% of total sales value**

A near-textbook Pareto split — confirms the ABC classification captured real, concentrated value, not an artifact of the threshold chosen.

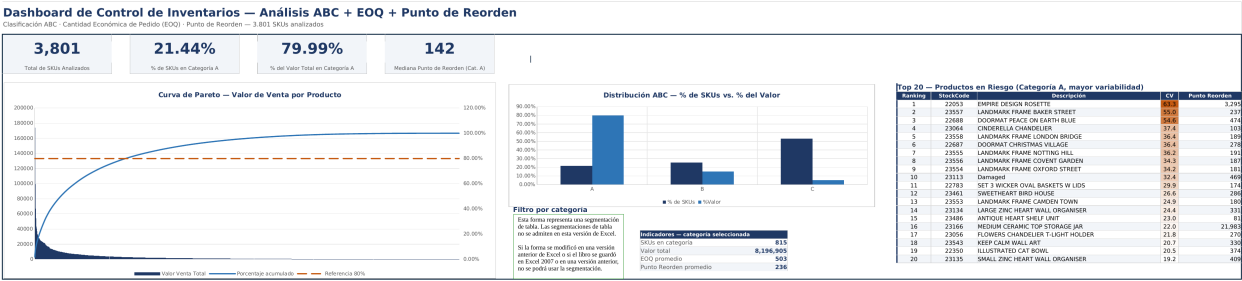
- **53.17% of products (Category C, 2,021 SKUs) generate only 5.00% of value**

More than half the catalog barely moves the needle in sales terms, and does not justify tight inventory control.

- **Median (142 units), not mean (235.6 units), used for the Category A Reorder Point KPI**

A single transaction of 74,215 units for one SKU (StockCode 23166) — nearly 260× its typical order line — inflated the average. The outlier was kept (no evidence of a data entry error) but documented as the reason the median better represents a typical Category A product.

## Dashboard



KPI summary, Pareto curve with 80% reference line, ABC value distribution, and the Top 20 highest-variability Category A products. Interactive in the source workbook via a category slicer.

Assumptions & limitations

- Order cost (£15/order), holding cost (20% of unit price), and lead time (7 days) are documented assumptions, not values present in the raw dataset.
- The dataset has no live inventory levels — it is historical sales data, so Reorder Point and EOQ are planning estimates, not a live stock-tracking system.
- 236 products (6.2%) have zero calculated standard deviation: 130 because they have only one recorded sale (too little data to estimate variability), and 106 because every recorded sale was for the exact same quantity (genuinely stable demand).
- The 80% ABC threshold, the 95% service level for safety stock, and the CV-based risk ranking are standard methodological choices, not the only valid ones.

Data source: Online Retail Dataset (Kaggle) — 541,909 raw transaction lines from a UK-based online retailer, cleaned to 3,801 unique products. Full workbook, methodology detail, and repository structure:  
[github.com/danielforero26/01-inventory-abc-eoq-dashboard](https://github.com/danielforero26/01-inventory-abc-eoq-dashboard)